Prediction-Based Adaptive Edge Caching Update Strategy for the Internet of Vehicles

Sirui Ruan
College of Computer Science
South-Central Minzu University
Wuhan, China
2024120427@mail.scuec.edu.cn

Rui Hou*
College of Computer Science
South-Central Minzu University
Wuhan, China
hourui@mail.scuec.edu.cn

Wei Li
College of Computer Science
South-Central Minzu University
Wuhan, China
liwei@mail.scuec.edu.cn

Yuanai Xie
College of Computer Science
South-Central Minzu University
Wuhan, China
xieyuanai@mail.scuec.edu.cn

Wanneng Shu
College of Computer Science
South-Central Minzu University
Wuhan, China
shuwanneng@whu.edu.cn

2026 IEEE/ACM International Symposium on Quality of Service (IWQoS) | 979-8-3195-1703-6/26/$31.00 ©2026 IEEE | DOI: 10.1109/IWQoS70441.2026.11661158

***Abstract*—The integration of IoV and MEC leads to a surge in delay-sensitive content requests. In viaduct scenarios, 3D structures cause frequent LOS/NLOS transitions, posing challenges to traditional edge caching. This paper presents an edge intelligence-based adaptive caching framework with LSTM popularity prediction and DQN decision-making. Experiments on real-world SUMO trajectories show that our method achieves 77.35% cache hit ratio and 31.2 ms average latency, which outperforms baseline schemes by up to 6.06% and 11.4%, respectively.**

***Index Terms*—Internet of vehicles, edge intelligence,prediction-based,adaptive cache,viaduct**

I. Introduction

The integration of Internet of Vehicles (IoV) and mobile edge computing (MEC) has led to a rapid growth of delay-sensitive content requests. Vehicular applications such as real-time navigation and safety services impose strict quality of service (QoS) requirements [1]. Edge caching at roadside units (RSUs) is an effective way to reduce latency [2] by deploying popular content near vehicles. However, dynamic vehicle mobility, time-varying content popularity, and heterogeneous channel conditions bring great challenges to efficient caching design.

These challenges are more critical in elevated highway scenarios, where three-dimensional infrastructure causes frequent transitions between line-of-sight (LOS) and non-line-of-sight (NLOS) links. Such elevation-induced heterogeneity cannot be handled by traditional models.To address these issues, this paper proposes an LSTM-DQN-based adaptive edge caching framework, which combines content demand prediction and dynamic caching decision with the full consideration of the 3D propagation characteristics of viaduct scenarios [3].

II. Proposed Method

A. Three-Dimensional Viaduct Model

As shown in Fig. 1, we consider an IoV-MEC scenario on a 1000-meter elevated highway with five segments: ground road, uphill ramp, bridge deck, downhill ramp, and ground road. Roadside Units (RSUs) with limited edge cache are deployed at fixed intervals to serve vehicles. To describe elevation-related propagation characteristics, we use a 3D Cartesian coordinate system (x, y, z), where x is the driving direction, y is the lane direction, and z is the height from the ground.

Based on the relative elevation $|\Delta z|$ between vehicles and RSUs, we classify communication links into three QoS types: LOS link ($|\Delta z| < 2$ m, 5 ms delay, 1% packet loss); Ramp link ($2 \leq |\Delta z| < 15$ m, 10 ms delay, 10% packet loss); NLOS link ($|\Delta z| \geq 15$ m, $\geq$25 ms delay, $\geq$30% packet loss). This elevation-aware classification provides differentiated weights for the DQN reward, enabling the caching policy to adapt to dynamic viaduct channel conditions.

B. LSTM-Based Content Demand Prediction

Vehicular content requests show strong temporal correlation. We use an LSTM network to predict short-term content popularity at each RSU. Let $r_f(t)$ be the request count of content f in time slot t. Given a window of length T, the historical request sequence is:

$$\mathbf{r}_f(t) = [r_f(t-T+1), \ldots, r_f(t)]. \tag{1}$$

The LSTM outputs the predicted request for the next slot:

$$\hat{r}_f(t+1) = \mathcal{F}_{\text{LSTM}}(\mathbf{r}_f(t); \theta), \tag{2}$$

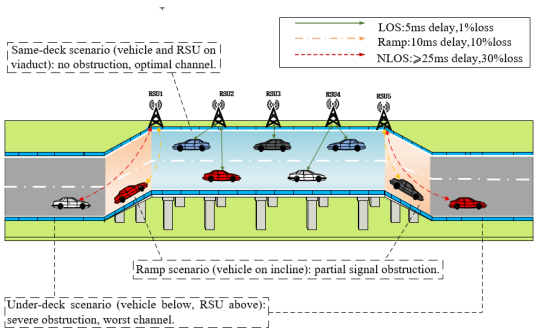


Fig. 1. Schematic diagram of the V2R scenario in the overpass

979-8-3195-1703-6/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: Universita degli Studi di Genova. Downloaded on October 23,2026 at 09:19:39 UTC from IEEE Xplore. Restrictions apply.

where θ denotes network parameters. The normalized popularity vector $\hat{\mathbf{p}}(t+1)$ serves as a key state feature for caching decisions.

C. DQN-Based Adaptive Cache Control

We formulate the dynamic caching problem as a Markov Decision Process (MDP) and solve it via a DQN agent. At time slot t, the system state is defined as:

$$\mathbf{s}(t) = \{\hat{\mathbf{p}}(t), \rho(t), \mathbf{q}(t), \boldsymbol{\eta}(t)\}, \tag{3}$$

where $\rho(t)$ is cache occupancy, $\mathbf{q}(t) = [q_{\text{delay}}(t), q_{\text{loss}}(t)]$ denotes real-time QoS metrics, and $\boldsymbol{\eta}(t)$ characterizes elevation-related link quality.

A composite reward function is designed to optimize multiple QoS goals:

$$R(t) = \alpha \cdot H(t) - \beta \cdot D(t) - \gamma \cdot L(t) - \delta \cdot B(t), \tag{4}$$

where $H(t)$, $D(t)$, $L(t)$ and $B(t)$ denote cache hit ratio, access delay, packet loss and backhaul bandwidth consumption, respectively. The weights of delay and loss penalties are adjusted adaptively by link type (LOS, ramp, NLOS) to reflect channel quality degradation in 3D viaduct scenarios.

D. Framework Overview

As shown in Fig. 2, the proposed edge intelligence framework consists of two coupled modules deployed at each RSU: content demand prediction and DQN-based caching control. At each decision interval, the RSU predicts future content popularity from historical requests. The predicted popularity and real-time network states are input to the DQN agent to derive optimal caching and replacement strategies, enabling QoS-aware adaptive edge caching under 3D viaduct environments.

III. Simulation Results

This section presents comprehensive performance evaluations of the proposed LSTM+DQN framework under the 3D elevated viaduct scenario. The simulation is conducted based on realistic SUMO vehicle trajectories, with 478 vehicles, over 50,000 trajectory points, a content library size $F = 100$, and an RSU cache capacity $C = 10$. We compare our scheme with five baselines: Random, LRU, LFU-DA, W-TinyLFU, and LSTM-CRP.

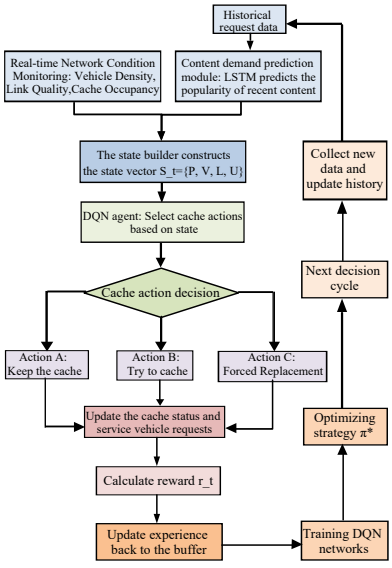
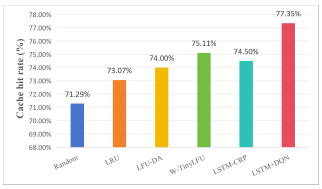
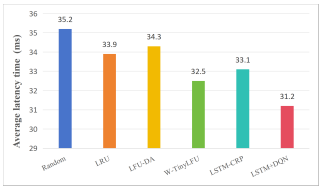


Fig. 2. Overview of the proposed caching framework

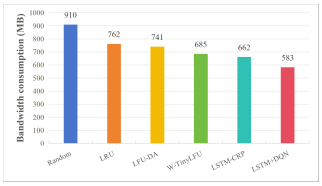
As shown in Fig. 3(a), the proposed LSTM+DQN framework achieves the highest cache hit rate of **77.35%**, outperforming all baseline schemes. Fig. 3(b) demonstrates that our approach reduces the average access latency to **31.2 ms**, which is the lowest among all evaluated strategies. In Fig. 3(c), the LSTM+DQN scheme significantly reduces backhaul bandwidth consumption to **583 MB**, effectively alleviating core network pressure. Fig. 3(d) shows that the proposed method maintains the highest cache utilization of **99.21%**, fully utilizing edge cache resources to improve user QoS.



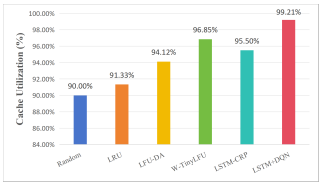
(a) Cache hit rate comparison.



(b) Average latency comparison.



(c) Bandwidth consumption.



(d) Cache utilization comparison.

Fig. 3. Performance comparisons of different caching strategies

IV. Conclusion

This paper proposes an LSTM+DQN-based adaptive edge caching framework for elevated viaduct scenarios. By explicitly modeling elevation-induced LOS/NLOS transitions and integrating prediction with reinforcement learning, the framework enables QoS-aware caching under dynamic vehicular conditions. Simulations on realistic SUMO trajectories show that the proposed method achieves 77.35% cache hit rate and 31.2 ms average latency, outperforming baseline schemes by 6.06% and 11.4%, respectively.

V. Acknowledgment

This work is partially supported by the National Natural Science Foundation of China, Grant Number 62372479, and the Leading Talents Program of the National Ethnic Affairs Commission of the People's Republic of China.

References

[1] L. Zhang, L. Zhao, Q. Wu *et al.*, "Vehicle-assisted edge caching strategy: integrating content freshness and user preferences," *J. Supercomput.*, vol. 81, no. 15, 2025.

[2] H. Wu, B. Wang, H. Ma *et al.*, "Collaborative caching relay algorithm based on recursive deep reinforcement learning in mobile vehicle edge network," *Ad Hoc Netw.*, vol. 152, p. 103313, 2024.

[3] C. Liu, Z. Xue, C. Liao *et al.*, "Drl-enhanced vehicular edge caching addressing content dynamics and complex intersections," *IEEE Internet Things J.*, vol. 12, no. 2, pp. 1732–1745, 2025.

Authorized licensed use limited to: Universita degli Studi di Genova. Downloaded on October 23,2026 at 09:19:39 UTC from IEEE Xplore. Restrictions apply.